

ABHIJIT GHOSH

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OBJECTIVE

Geometrical Assurance Engineer with daily hands-on experience in Siemens NX for design modifications and 3D modeling, CETOL for 3D tolerance stack-up / variation simulation analysis (VSA), and Teamcenter Visualization Mockup for cross-functional design reviews. Skilled in GD&T application and dimensional management within automotive product development, seeking to deliver robust geometrical solutions for vehicle cab assemblies.

PROFESSIONAL EXPERIENCE

Engineer – Geometrical Assurance Engineering (Stellantis)

May 2025 – Present

Infosys Limited

Mysuru, India

- Improved assembly yield from 54.34% to 99.90% per million units by performing 3D tolerance stack-up / variation simulation analysis (VSA) using CETOL integrated with Siemens NX, identifying variation risks and recommending critical tolerance optimization changes across BIW and interior assemblies.
- Ensured zero critical clashes across 15+ interface zones as validated through Teamcenter Visualization Mockup reviews and virtual clearance analysis, by evaluating gap, flush, parallelism, and alignment conditions against aesthetical and functional requirements for exterior and interior components.
- Executed design modifications and geometry corrections directly in Siemens NX as the primary modeling environment, ensuring updated 3D models reflected tolerance optimization recommendations and maintained assembly-level datum integrity (3-2-1 principle).
- Reduced project management cycle time by ~40% as measured by stakeholder feedback and task completion velocity, by architecting automated workflows using Microsoft Power Automate and Microsoft Lists that streamlined dimensional convergence tracking and enhanced data transparency with the client.
- Achieved first-pass design approval on 90%+ reviewed assemblies as tracked through gate review acceptance rates, by applying GD&T callouts (ASME Y14.5) with proper datum strategy during design reviews to maintain dimensional accuracy across sheet metal and plastic components.
- Contributed to physical-virtual correlation activities as measured by reduced deviation between simulation and CMM measurement reports, by reviewing inspection fixture data and correlating virtual tolerance models with production dimensional data to support dimensional convergence during build phases.

Systems Engineer Trainee

October 2024 – April 2025

Infosys Limited

Mysuru, India

- Validated component fit and alignment for 10+ design change orders as confirmed through design review acceptance, by performing 1D and 3D tolerance stack-up calculations and applying GD&T principles to verify datum targets and reference systems at part and subassembly levels.
- Accelerated design-to-manufacturing handoff as measured by reduction in engineering change requests during production readiness reviews, by executing 3D model modifications in Siemens NX and adding PMI annotations in CATIA V5 for BIW and interior components including sheet metal and plastic parts.
- Validated assembly-level clearances and component interfaces using Teamcenter Visualization Mockup as confirmed through clash-free design review sign-offs, by loading multi-CAD assemblies and verifying spatial relationships prior to physical prototyping.
- Identified 3 critical stress concentration zones before physical prototyping as validated through FEA simulation convergence, by performing structural analysis in Ansys Workbench to check load behavior and validate design assumptions against functional and mountability requirements.

Subject Matter Expert – Mechanical Engineering

July 2023 – September 2024

Vaidik Eduservices Pvt. Ltd

Remote, India

- Supported 500+ students across core mechanical engineering subjects as measured by content engagement and resolution rates, by developing structured problem-solving guides covering GD&T fundamentals, tolerance analysis, and manufacturing process knowledge for academic assessments.

Research Intern – Metallurgical & Materials Engineering

March 2023 – June 2023

IIT Jodhpur

India

- Contributed to published research on lithium-ion battery electrode performance as measured by successful material characterization milestones, by conducting experimental analysis on novel in-situ volume contractible metal halide negative electrodes using statistical data analysis and laboratory measurement methods.

SKILLS

GAE / Dimensional Mgmt	3D Tolerance Stack-Up Analysis, Variation Simulation (VSA), GD&T (ASME Y14.5), Gap & Flush Analysis, Datum Strategy (3-2-1 Principle), Dimensional Convergence, Physical-Virtual Correlation
Core CAD / Simulation	Siemens NX (primary – daily use for design modifications & 3D modeling), CETOL (3D tolerance analysis), Teamcenter Visualization Mockup (assembly reviews), CATIA V5, Ansys Workbench, PMI (Product & Manufacturing Information)
PLM & Data	Siemens Teamcenter, NX PLM, SQL
Measurement & Quality	CMM Fundamentals, Check Fixtures, Inspection Reports, SPC, Cp/Cpk
Domain Knowledge	BIW, Exterior & Interior Components, Sheet Metal & Plastic Manufacturing, Automotive Product Development
Programming & Tools	Java, Microsoft Power Platform, Power Automate, MS Office
Languages	English, German (Elementary Proficiency)

EDUCATION

Bachelor of Technology in Mechanical Engineering — <i>Techno Main Salt Lake</i>	July 2022 DGPA: 8.55
Diploma in Mechanical Engineering — <i>Santiniketan Institute of Polytechnic</i>	June 2019 DGPA: 7.8

CERTIFICATIONS

- Certified Lean Six Sigma AI Yellow Belt – Sparen & Gewinn Consulting (CSSC Accredited, Aug 2026)
- NX PLM-1 Engineering Author – Infosys (May 2026)
- Fundamentals of Microsoft Power Platform – Infosys (May 2026)
- Teamcenter 14.1x Visualization Mockup – Tata Technologies (Jan 2026)
- Fundamentals of Finite Element Analysis – Tata Technologies (Jan 2026)
- Product & Manufacturing Information (PMI) – Tata Technologies (Dec 2025)
- Infosys Certified GD&T Overview
- Infosys Certified Introduction to NX

AWARDS

- **Infosys Insta Award** (Feb 2026): Recognized by leadership for outstanding contributions to project efficiency through custom workflow automation initiatives.
- **High Performer** in Infosys Foundation Program.